

Project Changes & Additions

Automatic Timetable Generator | Django + Google OR-Tools CP-SAT
4 October 2026 | Baseline commit **9f4c449**

Metric	Before	After
Application routes	15	30
Python modules	12	19
Test cases	59	68
Test result	—	68 passed, 0 failed
Known open bugs	7	0

1. Executive Summary

Since the last commit the project gained **6 new features** (~3,700 lines across 25 new files) and grew by **~2,845 insertions** across 15 existing files. A full audit of every page and form uncovered **7 real bugs**, all of which are now fixed and covered by regression tests.

2. New Features Added

2.1 PDF Import — scheduler/pdf_import.py (975 lines)

- Imports a college timetable from a real-world PDF rather than requiring hand-built JSON.
- Parses ruled tables with selectable text using **pdfplumber**.
- Tolerates deliberately messy documents: shuffled column order, **DD Mon YYYY** dates, working days expressed as a range, the lab flag written as the letter **P**, implied rooms, and 12 numbered divisions.
- Two-stage flow.** Stage 1 parses the PDF and reports how each table was recognised. Stage 2 lets the administrator correct the column mapping, then performs the exact same validated atomic import the JSON path uses.
- Runs the identical validation as the JSON importer, so a bad PDF cannot corrupt data.

Route: /upload/pdf/ · **Template:** upload_pdf.html (208 lines)

2.2 PDF Export — scheduler/pdf_export.py (445 lines)

Four printable reports rendered with **reportlab**:

Report	Route
Full institution grid	/export/full.pdf
Faculty teaching load	/export/faculty.pdf
Scheduling issues	/export/issues.pdf
Single division	/export/division/<id>.pdf

2.3 Recommendations Engine — scheduler/recommendations.py (514 lines)

- Diagnoses every unresolved scheduling issue and proposes a concrete remedy.

- **Each remedy is trial-solved before being shown**, so the claimed hour gain is verified rather than guessed.
- Multiple remedy types: add a room, raise capacity, move a teacher, rebalance load.
- Identical remedies are de-duplicated — one undersized lab no longer produces three identical proposals.

2.4 Approval Workflow — *scheduler/proposals.py* (190 lines)

- Proposals remain **PENDING** until an administrator explicitly approves them.
- Nothing is ever changed automatically.
- Approval applies the change inside a transaction, re-solves, and reports the real before/after hours and the actual issue count resolved.
- Rejection records who declined it and why.

Route: /proposals/ · **Template:** proposals.html (116 lines)

2.5 Week Preferences — *scheduler/preferences.py* (274 lines)

- Set working days, start/end time, period length and lunch break.
- **Analyses the proposed week shape before saving** — if it cannot hold the curriculum, the change is refused with the specific reasons rather than applied blindly.
- Per-division overrides (first/last period, consecutive lab blocks).
- Warns when a shared period grid change damages another semester.

Routes: /preferences/, /preferences/division/<id>/

2.6 Notifications — *scheduler/notifications.py* (256 lines)

- Teachers and students are told when **their** timetable changes, with a before/after diff of what moved.
- Administrators are told when issues appear and when proposals await approval.
- One indexed unread count per page for the navigation badge, via context_processors.py (14 lines).

Route: /notifications/ · **Template:** notifications.html (112 lines)

2.7 Supporting Additions

File	Lines	Purpose
scheduler/capacity.py	184	Reports facility shortfalls before solving
seed_demo_accounts.py	149	One admin, teacher and student demo login
templatetags/scheduler_tags.py	15	Template filters for the grid
migrations/0004_...py	276	9 new models
migrations/0005_...py	31	SET_NULL on proposal links
7 new templates	1,068	Capacity, issues, preferences, PDF upload, ...

2.8 Dependencies Added

```
pdfplumber>=0.11.0 # PDF import: ruled tables from a timetable PDF
reportlab>=4.0.0   # PDF export: division, faculty-load and issue reports
```

3. Changes to Existing Files

15 tracked files changed — **2,845 insertions, 353 deletions.**

File	Change	What it does
scheduler/views.py	+754	13 new views; 15 → 30 routes
scheduler/solver.py	+593	Backtracking room matching; records issues
scheduler/tests.py	+743	59 → 68 test cases
scheduler/models.py	+378	9 new models
scheduler/importer.py	+84	Shared by JSON and PDF import paths
scheduler/urls.py	+25	New route table
templates/base.html	+127	Navigation, badge, confirm dialog
templates/dashboard.html	+89	Capacity and issue metrics
templates/timetable_view.html	+20	Room and filter display
templates/upload_json.html	+20	Validation error reporting
README.md	+212	Full rewrite
PROJECT_EXPLAINED.md	+146	Architecture notes
requirements.txt	+6	Two new dependencies
settings.py	+1	Context processor registration
db.sqlite3	binary	Rebuilt demo data

4. Bugs Found and Fixed

A full audit exercised all 30 routes as administrator, teacher, student and anonymous user, plus every form, the real JSON and PDF importers, and all PDF exports. **Seven genuine defects were found.**

Bug 1 — Two features crashed on first use (missing imports)

SchedulingPreference was used in views.py but never imported; generate_time_slots was used in preferences.py but never imported. The preferences page raised NameError on every save. Found with pyflakes.

Bug 2 — “Apply preferences” always crashed

preferences.py saved the *candidate* object built from the submitted form — an unsaved duplicate — so it tried to INSERT a second row for a semester that already had one:

```
IntegrityError: UNIQUE constraint failed:
  scheduler_schedulingpreference.semester_id
```

Now updates the stored row.

Bug 3 — Week shape never re-timed existing periods

generate_time_slots used get_or_create, so an existing period kept its original start and end time. Shrinking the college day to 09:00–15:00 left a period starting at **15:00 on a day that ended at 15:00**. The published grid silently disagreed with the preference that produced it. Now changed times are refreshed.

Bug 4 — Blank time field returned HTTP 500

Clearing the start time field produced ValueError: invalid literal for int() with no handler, so the whole page 500'd. Now raises a proper validation error rendered as a readable form message.

Bug 5 — Administrator got a 500 on the personal timetable page

my_timetable_view assumed every signed-in user has either a teacher or a student profile. An administrator has neither, so notification links pointed them at a crash. Now redirected to their own landing page.

Bug 6 — Approving a proposal deleted it mid-approval

ProposedChange.issue used on_delete=CASCADE. Approval re-runs the solver, which deletes and recreates every SchedulingIssue row — taking the proposal being approved with it. The final save then failed:

```
ProposedChange.NotUpdated:
  Save with update_fields did not affect any rows.
```

The change applied, but the page returned an error. Changed to SET_NULL, so a pending proposal survives a re-solve.

Bug 7 — Identical proposals and triple notifications

One undersized laboratory produced three identical “Add a 70-seat laboratory” proposals — one per affected issue — and six notifications instead of two. Added signature-based de-duplication; a single approval now resolves every issue that shared the remedy.

Also corrected

- notify_timetable_changed passed composite (assignment_id, slot_code) tuples into an id__in lookup. This crashed *after* the transaction committed, so the fix applied but the user saw an error page.
- A stale test asserted a navigation label that no longer existed.

5. Testing

Test cases	68 (59 existing + 9 new regression tests)
Result	68 passed, 0 failed
Runtime	~253 seconds
Static analysis	pyflakes clean — no undefined names
Route coverage	30 of 30 routes exercised in all 4 roles

Security verification. Access control was checked exhaustively. **Zero permission leaks:** all 17 administrator-only routes correctly redirect teachers, students and anonymous users. Open-redirect protection on the login form confirmed working. Logout is POST-only.

Regression tests added

Each new test names the symptom it prevents, so the defect cannot silently return:

- **1.** Administrator without a profile is redirected, not crashed
- **2.** A proposal survives the re-solve its own approval triggers
- **3.** Applying preferences updates rather than inserts
- **4.** Changing the week shape re-times existing periods
- **5.** Blank time field yields a form error, not a 500
- **6.** Blank period length yields a form error, not a 500
- **7.** Identical remedies are proposed once
- **8.** One approval clears every issue sharing that remedy
- **9.** Recommending twice does not duplicate pending proposals

6. Demo Accounts

Role	Username	Password	Lands on
Administrator	admin	admin123	Dashboard, solver, management
Teacher	teacher.demo	Teach@12345	Personal teaching timetable
Student	student.demo	Study@12345	Personal class timetable

Change these passwords before any real deployment.

Current demo dataset: *Fall Semester 2026* — 7 divisions, 11 subjects, 17 rooms, 7 teachers, 20 assignments, 42 periods across 6 days. The solver schedules **64 of 64 requested hours with 0 unresolved issues**.

7. Known Limitation

TimeSlot is global rather than per-semester, because the code treats the period grid as an institution-wide shared structure. Importing a second timetable therefore permanently alters the grid for all semesters. The preferences flow detects this and warns which other semesters were affected, but if true multi-semester operation is needed, TimeSlot should become semester-scoped.

8. Repository Status

All work is **uncommitted**. Nothing has been pushed. `db.sqlite3` is modified and contains the rebuilt demo dataset; consider whether it belongs in version control.

```
Modified: 15 tracked files
New:      25 untracked files
```